

PRACTICAL EXAM TRAINER GUIDE

WATER & PREVENTIVE MAINTENANCE FOUNDATION

COFFEE TECHNICIANS PROGRAM

實作測驗 講師指南

水&預防性維護基礎課程

咖啡技師學程

This practical exam simulates tasks which are commonly performed during a preventive maintenance visit. These tasks are divided into four (4) Sections. To receive full credit for each section the learner must record all relevant observations, calculations, and task completions requested in the Practical Exam Learner Worksheet.

本實務測驗模擬在預防性維護檢測時常見的項目。這些項目分為四個部分。要獲得每一個部分的學分，學員必須記錄下實務測驗學員工作表中所要求的所有相關觀察、計算和任務完成情況。

IMPORTANT SAFETY NOTICE: 重要安全事項：

This exam includes activities which involve the learner interacting with hand tools. It is the sole responsibility of the AST to ensure that:

本測驗包含讓學員操作手動工具的環節。SCA 講師需負責確保下列事項：

All Learners are properly supervised at all times while working with hand tools

- 所有學員隨時都在適當的監督下操作手動工具。

An easily accessible, fully-equipped first aid kit is on site

- 現場備有完整配備且方便取用的急救箱。

All Learners use appropriate Personal Protective Equipment during course activities

- 所有學員在課程活動期間皆使用適當的個人防護配備。

All working stations are set up to meet the safe workplace recommendations detailed in guidebook sections 1.01 and 1.06 and all local workplace safety standards

- 所有工作站的設置皆須達到指引手冊章節 1.01、1.06 所載明之安全建議及所有當地工作場域安全標準。

If unsafe behavior is observed at any time, the exam is stopped for the Learner and a score of zero (0) is to be given for that section of the exam

- 無論何時，當注意到學員做出危險舉動時，該學員的測驗立即中止，並於該部分的測驗給予(0)分。

Learners exhibiting unsafe behavior are provided with feedback and demonstrate an understanding of safe behavior before attempting a retake

- 須對學員的危險舉動給予回饋，並在嘗試重考前確認學員對安全行為的理解。

Learners are familiar with the safe use of all hand and measurement tools to be used prior to starting this exam

- 在開始本測驗前，學員需要熟悉所有手動工具及測量工具的安全使用方式。

All course activities are conducted in accordance with local safety regulations and best practices

- 所有課程活動皆按照當地安全規定及最佳實務進行。

All stations which involve live electricity are connected in accordance with local electrical code and workplace safety regulations

- 所有需用電之工作站皆須依照當地電工標準及工作場域安全規定進行連接。

All stations which involve live electricity are connected to power using GFCI devices which have been tested for proper function before beginning this exam.

- 所有需用電之工作站連接到電源皆須使用漏電斷路設備，並在測驗前通過功能測試。

All outlets and devices are properly grounded and tested by the AST to confirm proper grounding before beginning this exam

- 所有插座及設備在測驗前皆確實接地並經由講師確認。

All hand tools and measurement devices are inspected for safe and proper function before beginning this exam

- 所有手動工具及測量設備在測驗前皆經測試後確認安全性及功能正常。

If local electrical code or workplace safety regulations do not allow for a prescribed activity, the AST is not to conduct this activity and is to contact SCA Education to discuss alternative activities.

假如當地安全電工標準或工作場域安全規定不允許進行指定的活動，講師將不進行該活動並連絡 SCA 教育團隊討論替代方案。

Section 1 – Measuring Water Composition

第一章 - 測量水的組成

- Time Available: 10 minutes for all three (3) parts
- Points Available: 15, 11 points required to pass
- Silent working is required.
- This section is designed to evaluate the Learner's ability to:
 - Measure the Total Hardness and Alkalinity of a water sample
 - Record Total Hardness and Alkalinity measurements on the SCA Water Chart
 - Use an Electrical Conductivity meter to measure the estimated TDS of a water sample

測驗時間：總共十分鐘，分三個部分

得分：15分，通過門檻為11分

作業中禁止交談

本章節旨在評估學員下列能力：

測量水質樣本的總硬度及鹼度。

將總硬度及鹼度的測量結果記錄在SCA水質圖上。

使用導電度計測量水質樣本的TDS概略值。

Station Preparation 工作站前置作業

Create a space for individual work in this exam, either by erecting visual dividers between learners or placing them in separate rooms.

- 利用在學員間設置視線擋板或是分配至獨立工作間以隔離出本測驗的個人工作空間。準備兩個大Prepare 2 different approximately 500 mL samples of water for evaluation by the learners:

- 約 500ml 的不同水質樣本供學員們評估

Label each sample with a 3-digit number

- 以3位數字為各個水質樣本做樣本編號

Samples must have the following parameters:

- 水質樣本必須符合下列參數：

- Total Hardness between 30 and 160 ppm CaCO₃
- Alkalinity between 30 and 160 ppm CaCO₃
- Samples must differ from each other on both parameters by at least 20 ppm

CaCO₃ 總硬度介於 30~160ppm CaCO₃ 之間

鹼度介於 30~160ppm CaCO₃ 之間

水質樣本之間在兩個參數上皆須有 20ppm CaCO₃ 的差異

- Samples may be prepared using prepared coffee mineral mixes, AST-mixed concentrate solutions, or bottled water. See The SCA Water Quality Handbook section __ for guidance

水質樣本的備製可使用調製好的咖啡礦物質調和物、AST 調和濃縮液或是瓶裝礦泉水。

可參閱 SCA 水質手冊第 __ 章節尋求指引。

Provide drop-count Total Hardness and Alkalinity test kits, electrical conductivity TDS meters, and necessary sampling and testing accessories and supplies

- 提供測驗總硬度及鹼度的滴定試劑、導電度 TDS 計、必要的採樣及測試器材。

Provide printed copies of the SCA Water Chart, 1 per student minimum

- 提供 SCA 水質圖的影本，至少每位學生一份。

Provide a visual reference for the student to use during this section of the exam which indicates the color change expected for each drop-count test and the conversion factor between drops required for color change and ppm CaCO₃

- 提供標示出各個滴定試驗所預計的顏色變化，以及顏色變化所需的滴定量和 ppm CaCO₃ 的換算係數之視覺參照，供學生於本章節測驗中使用。

Provide each Learner with a calculator

- 提供各個學員一台計算機。

Provide each Learner with a clipboard.

- 提供各個學員一塊書寫夾板。

Trainer Instructions 講師指示

Review the Learner Instructions of the exercise with the Learner.

- 陪同學員詳讀學員測驗指示

Assign a water sample to the Learner for evaluation. A sample may be evaluated by more than one learner. If a retake is required for this section, assign a different sample to the Learner for their second attempt.

- 分配水質樣本供學員評估。一個水質樣本可能由多個學員進行評估。如果本章節需要重考，請在第二次嘗試時為學員分配一個不同的水質樣本。

If the Learner is unable to obtain Total Hardness and Alkalinity measurements in the Skill A portion of this section, provide values for Total Hardness and Alkalinity within the SCA Core Zone for their use in Skill B.

- 如果學員在本章節的技能 A 部分，無法獲得總硬度及鹼度測量值，請提供位於 SCA 水質核心區的一組總硬度及鹼度數值予學員們用於技能 B 部分的測驗。

Alert the Learner to begin and start the timer.

- 提醒學員測驗開始並開始計時。

The Learner is to alert you when they have finished.

- 學員將會在完成測驗時告知你。

Once the Learner has announced that they have finished, they cannot add to or change their answers. If time expires, the Learner cannot add to or change their answers.

- 一旦學員宣布他們完成測驗時，他們將無法增加或修改答案。如果時間結束時，學員亦無法增加或修改答案。

Silent work is required. If a Learner compromises the integrity of the exam for another, you must reset the exam for the next Learner.

- 作業中禁止交談。如果有學員破壞了別的學員考試之完整性，請為下一位學員重新設置測驗。

Learner Instructions 學員指示

A designated water sample has been prepared for your use in this section of the exam. You are to measure several parameters of this sample and record your results.

- 指定的水質樣本已準備好以供本章節測試中使用。你將需要測量該水質樣本的幾個參數並記錄結果。

There are three (3) parts to this section of the exam.

- 本章節測驗一共有 3 個部分。

Read the instructions for each part thoroughly.

- 請詳閱各個部份的說明。

The trainer will tell you when your time begins.

- 計時開始時，講師將會告知你。

Alert your trainer when you have completed this section.

- 完成本章節測驗時請通知你的講師。

Once you have alerted the trainer, or if time expires, you will not be able to change your answers.

- 一旦你通知講師或是時間結束，你將無法再修改答案。

Skill A - Measure the Total Hardness and Alkalinity of a water sample

技能 A -測量水質樣本的總硬度及鹼度

Use the provided test kit to measure the Total Hardness and Alkalinity of the provided sample. You may reference the visual aid which the AST has provided.

- 使用提供的試劑測量水質樣本的總硬度及鹼度。你可以參考講師所提供的視覺參照。

Record your result on the Sample Preventive Maintenance Checklist.

- 請將你的結果記錄在預防性維護核對表。

Skill B - Record Total Hardness and Alkalinity measurements on the SCA Water Chart

技能 B-將總硬度及鹼度測量結果記錄在 SCA 水質圖

A copy of the SCA Water Chart has been provided for your use in this section.

- 本章節提供了一份 SCA 水質圖影本供你使用。

Perform any calculations which may be required to convert the measurements you obtained in Skill A of this section into ppm CaCO₃. You may reference the visual aid which the AST has provided.

- 將你在本章節技能 A 所得之測量值，執行所需的運算轉換成 ppm CaCO₃ 的值。你可以參考講師提供的視覺參照。

If you were unable to obtain measurements in Skill A, your AST will provide values for your use.

- 如果你沒有在技能 A 取得測量值，講師將會提供數值予你使用。

Plot the Total Hardness and Alkalinity values you obtain on this chart.

- 在水質圖上標示你獲得的總硬度及鹼度值。

Skill C - Use an Electrical Conductivity meter to measure the estimated TDS of a water sample

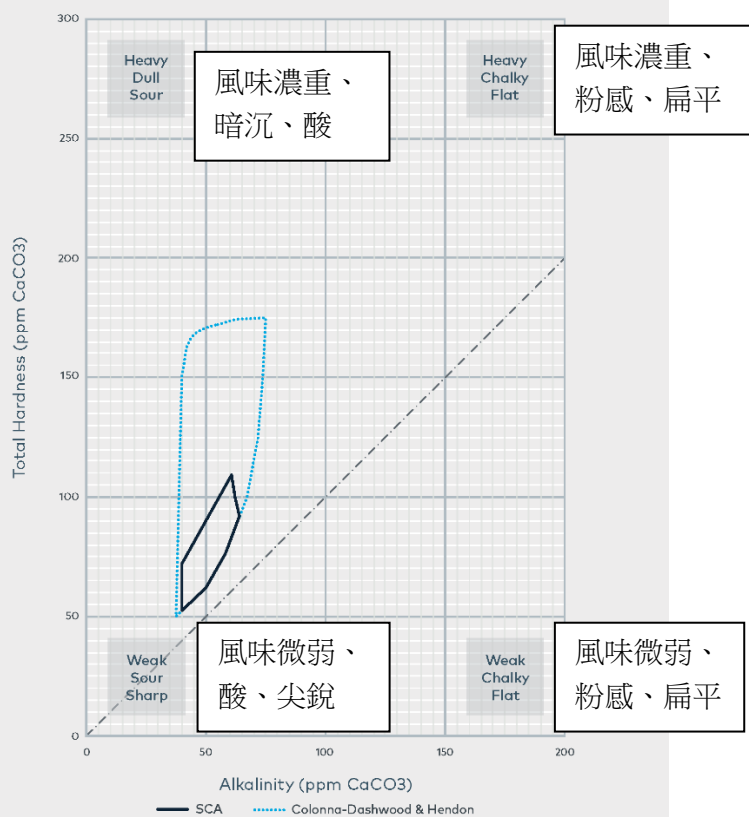
技能 C-使用導電度計測量水質樣本的 TDS 概略值。

Use the provided Electrical Conductivity meter to measure the estimated TDS of the provided water sample.

- 使用提供的導電度計測量水質樣本的 TDS 概略值。

Record your results on the Sample Preventive Maintenance Checklist

- 請將你的結果記錄在預防性維護核對表。



Scoring 計分

Place the resulting sub-total score for each requirement in the corresponding table on the Practical Exam Learner Worksheet.

將得出的小計總分填入實務測驗學員工作表的對應表格

Skill 1A - Measure the Total Hardness and Alkalinity of a water sample

技能 1A-測量水質樣本的總硬度及鹼度

Task 任務	Points Possible 可能得分
Learner properly collects samples and fills testing vials 學員正確地收集水質樣本並裝滿試瓶。	1
Learner uses appropriate technique for adding reagent drops and agitating sample 學員使用適當的技巧添加滴定試劑並攪拌水質樣本。	1 per vial 每瓶 1 分
Learner correctly recognizes color change and drops added for Total Hardness 學員正確地辨認測量總硬度時的顏色變化及添加的滴定量。	1
Learner correctly recognizes color change and drops added for Alkalinity 學員正確地辨認測量鹼度時的顏色變化及添加的滴定量。	1
Learner correctly records measurements on Learner Worksheet 學員正確地將測量結果記錄在學員工作表。	1
Total 總分	6

Skill 1B - Record Total Hardness and Alkalinity measurements on the SCA Water Chart

技能 1B-在 SCA 水質圖記錄總硬度及鹼度測量值

Task 任務	Points Possible 可能得分
Learner correctly performs calculation to convert measurements into ppm CaCO ₃ 學員正確地執行運算將測量結果轉換成 ppm CaCO ₃ 值。	1 per value 每個數值 1 分
Learner accurately plots the Total Hardness result obtained on the provided chart 學員準確地在提供的圖表中標示出所獲得的總硬度結果。	2
Learner accurately plots the Alkalinity result obtained on the provided chart 學員準確地在提供的圖表中標示出所獲得的鹼度結果。	2

	Total 總分	6
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Skill 1C - Use an Electrical Conductivity meter to measure the estimated TDS of a water sample

技能 1C-使用導電度計測量水質樣本的 TDS 概略值。

Task 任務	Points Possible 可能得分	
Learner properly collects water sample 學員正確地收集水質樣本。	1	
Learner uses appropriate technique for using electrical conductivity meter 學員使用適當的技巧操作導電度計。	1	
Learner correctly records estimated TDS measurement on Learner Worksheet 學員正確地將 TDS 測量結果記錄在學員工作表。	1	
Total 總分	3	

Section 2 - Water Treatment System Maintenance

第二章-水處理系統維護

- **Total Time Available: 10 minutes**
- **Points Available: 13, 9 points required to pass**
- **Silent working is required.**
- **This section is designed to evaluate the Learner's ability to:**
 - **replace a cartridge in a drop-in style water filter system**

測驗時間：共十分鐘

得分：13分，通過門檻為9分

作業中禁止交談

本章節旨在評估學員下列能力：

更換置入式濾水器系統中的濾芯。

Station Preparation 工作站前置準備

- Create a space for individual work in this exam, either by erecting visual dividers between learners or placing them in separate rooms.
利用在學員間設置視線擋板或是分配至獨立工作間以隔離出本測驗的個人工作空間。
- Prepare a drop-in style water filter system and connect to water supply:
準備一組置入式濾水器系統並連接至供水。
 - System must be connected to either a building's plumbing or bottled water pumps using a readily accessible shut-off valve.
系統必須連接到自來水或是瓶裝水泵浦，使用方便使用的止水閥。
 - Filter system should be securely mounted in reasonable proximity to the water supply.
過濾系統須被牢固地安裝在合理靠近供水的地方。
 - A cartridge of any type must be installed in the filter housing
每個類型的濾芯皆須被安裝在濾水器外殼中。
 - Provide one "new" filter cartridge for each Learner. The cartridge need not be unused but must be dry.
為每位學員提供一組 " 新的 " 濾芯。該濾芯不需要是未被使用過的，但必須是乾燥的。
 - The filter system outlet must be connected to a shut-off valve for use in flushing the filter after replacement. This connection may utilize a tee if the filter system is also connected to a piece of coffee equipment.
過濾系統出水口須被連接到止水閥，在更換濾芯後用於沖洗濾水器。假如過濾系統還連接到別台咖啡設備，該接頭可以使用三通管
- Provide a filter housing wrench, container to capture water during the flushing process, and all materials required for replacing the filter cartridge.
提供一只濾水器外殼扳手，沖洗過程中用於盛水的容器，以及所有更換濾芯所需的材料。

Trainer Instructions 講師指示

- Review the Learner Instructions of the exercise with the Learner.
陪同學員詳讀學員測驗指示

- **Alert the Learner to begin and start the timer.**
通知學員測驗開始並開始計時
- Observe and evaluate their work and score the task according to the table below.
觀察並評估他們的作業情況並根據下列表格對任務進行評分。
- **Alert the Learner to begin and start the timer.**
通知學員測驗開始並開始計時
- The Learner is to alert you when they have finished.
學員將會在完成時告知你。
- If time expires prior to the completion of the task, the learner may receive partial credit for the work that has been completed.
假如時間在完成任務前結束，學員可以從已經完成的作業中得到部份分數。

Learner Instructions 學員指示

- A drop-in cartridge style water filtration system which is connected to a water supply has been prepared for your use in this section of the exam.
在本章節測驗中，已準備一組連結上供水的置入式濾水器系統供你使用。
- Service records for this filter system indicate that this cartridge was installed one year prior to today's date. Record this date in the appropriate space on your Learner Worksheet.
本過濾系統的維護紀錄顯示該濾芯是在今日日期的一年前安裝的。將該日期記錄在學員工作表適當的位置中。
- The trainer will tell you when your time begins.
你的測驗時間開始時講師將會告知你。
- You are to perform all necessary steps to replace the filter cartridge and return the system to service.
執行全部所需的步驟，完成濾芯的更換並讓系統恢復運作。
- When you have completed this section, mark the task as completed on your Learner Worksheet and alert your trainer.
當你完成本章節時，在你的學員工作表上註記該任務已完成並通知講師。

Scoring 計分

Place the resulting sub-total score for each requirement in the corresponding table on the Practical Exam Learner Worksheet.

將得出的小計總分填入實務測驗學員工作表的對應表格

Replace a cartridge in a drop-in style water filter system

更換置入式過濾系統的濾芯

Task 任務	Points Possible 可能得分
Learner turns off water supply to water filter 學員關閉濾水器的供水。	2
Learner removes filter housing and old water filter cartridge 學員拆下濾水器外殼並移除老舊濾芯。	2

Learner properly cleans and sanitizes filter housing 學員正確地清潔並消毒濾水器外殼。	2
Learner installs new water filter cartridge and reassembles housing properly 學員安裝新的濾芯並正確地重新組裝外殼。	2
Learner turns on water supply to filter and flushes system appropriately 學員打開濾水器的供水並適當地沖洗過濾系統。	2
Filter system does not leak 過濾系統不會漏水。	2
Learner records last change date and replacement on Learner Worksheet 學員在學員工作表上記錄最後更換日期及更換項目。	1
Total 總分	13

Section 3 - Coffee Brewer and Espresso Grinder Preventive Maintenance

第三章-咖啡沖煮機及義式磨豆機預防性維護

- Total Time Available: 20 minutes for both (2) parts.
- Points Available: 29, 20 points required to pass
- Silent working is required.
- This section is designed to evaluate the Learner's ability to:
 - Inspect and clean a **coffee batch brewer** level probe
 - Replace the burrs in an espresso grinder

測驗時間:兩個部份各 20 分鐘

得分:29 分，通過門檻為 20 分

作業中禁止交談

本章節旨在評估學員下列能力:

檢查並清潔滴濾式咖啡機的水位探針。

更換義式磨豆機的刀盤。

Station Preparation

工作站前置準備

- Create a space for individual work in this exam, either by erecting visual dividers between learners or placing them in separate rooms. One learner should not be able to see the work of a fellow learner.
利用在學員間設置視線擋板或是分配至獨立工作間以隔離出本測驗的個人工作空間。
學員不應能夠看見鄰近學員的作業情況。

For Skill A

在技能 A

- Prepare a coffee batch brewer with a removable water level probe.

準備一台配有可拆卸水位探針的滴濾式咖啡機

- This brewer **must not** be connected to electrical power
該咖啡機不得接上電源。
- The plug end of the power cord must be visible for the Learner to verify that it is unplugged.
電源線的插頭端必須在學員的視線內以確保未被插上。
- The brewer may, at the trainer's discretion, be connected to a water supply and the tank may be full of water.
根據講師的判斷，該咖啡機可以接上供水並在水箱裝滿水。

- Provide all tools and supplies needed to remove and clean level probe, possibly including:

提供拆卸及清潔水位探針所需的所有器材，可能包含:

- Screwdriver
螺絲起子
- Wrench
扳手
- Pliers
- Utility knife
- Steel wool or scrubbing pad

鉗子

工具刀

鋼絲絨或菜瓜布

For Skill B

在技能 B

- Prepare a fully-operational espresso grinder and matching electrical outlet.
準備一台可完整運作的義式磨豆機及合適的電源插座
 - This grinder **must not** be connected to electrical power
該磨豆機不得接上電源
 - The plug and outlet must be accessible so the Learner may plug it in to confirm function after completing the test activities.
插頭及插座必須方便取用，讓學員能在完成測驗項目後插上以確認功能。
- Provide all tools and supplies needed to access and remove grinding burrs, possibly including:
提供檢查及拆除刀盤所需的所有器材，可能包含：
 - Screwdrivers
螺絲起子
 - Wrench
扳手
 - Stiff-bristled brush
硬毛刷
 - Steel wool or scrubbing pad
鋼絲絨或菜瓜布
 - Appropriate lubricant
合適的潤滑劑

Trainer Instructions 講師指示

- Review the Learner Instructions of the exercise with the Learner.
陪同學員詳讀學員測驗指示。
- Alert the Learner to begin and start the timer.
通知學員測驗開始並開始計時。
- The learner will alert you when they have reached certain stages of the task. Evaluate their work and score the task according to the table below. Stop the timer while work is being evaluated.
學員每完成部份階段的任務時將會通知你。評估他們的作業情形並根據下表進行評分。
作業評量中暫停計時。
- If time expires prior to the completion of the task, the learner may receive partial credit for the work that has been completed.
如果時間在任務完成前結束，學員可以從已完成的作業中得到部份分數。
- To receive full credit, the learner must perform the tasks correctly and mark all tasks correctly in the Learner Practical Exam Worksheet
要獲得滿分，學員必須正確地執行任務並在實務測驗學員工作表標記出來

- This section includes working with tools. The learner must demonstrate safe working practices at all times during this section. Any unsafe actions will result in a score of 0 points for this section.
本章節包含工具操作。在測驗中學員必須隨時展現出安全的工作實踐。出現任何不安全的行為，本章節將會被評為 0 分。

Learner Instructions 學員指示

- There are two (2) parts to this section of the exam.
本章節測驗包含(2)個部分。
- Read the instructions for each part thoroughly.
請詳閱每個部份的說明。
- The trainer has provided a coffee batch brewer and fully-operational espresso grinder.
講師已提供一台滴濾式咖啡機及能完整運作的義式磨豆機。
- Service records for this equipment indicate that they were last serviced one year prior to today's date.
該設備的維護紀錄顯示最後維護日期為今日起算一年以前。
- In this section, you are to demonstrate a series of skills, detailed below, to complete preventive maintenance service on the provided batch coffee brewer and espresso grinder.
在本章節中，你需要展示下文所述的一系列技能，來完成對提供的滴濾式咖啡機及義式磨豆機的預防性維護服務。
- Record each task as complete on the Learner Practical Exam Worksheet, including observations where required.
在學員實務測驗工作表上將各個任務記錄為完成，並記錄下要求的各項觀測結果。
- The trainer will tell you when your time begins.
當你的測驗時間開始時，講師將會告知你。
- Alert your trainer when you have completed this section.
當你完成這個本章節時請通知講師。
- **This section includes working with tools. Safe working practices must be demonstrated at all times during this section. Any unsafe actions will result in a score of 0 points for this section.**
本章節包含工具操作。在測驗中必須隨時展現出安全的工作實踐。
任何不安全的行為將造成本章節被評為 0 分。

Skill A - Inspect and clean a coffee batch brewer level probe

技能 A-檢查並清潔滴濾式咖啡機的水位探針。

- The necessary tools are at this station.
所需的工具備妥在工作站中。
- Use the provided tools to:
使用提供的工具完成下列事項:
 - Remove any machine panels required to access the level probe.
拆除任何所需的機械面板以便檢查水位探針。
 - Remove the batch brewer level probe.
拆除滴濾式咖啡機的水位探針。
 - Inspect the probe for scale build-up and condition.
檢查探針上的結垢及探針的狀況。
 - Clean to remove any scale build-up
進行清潔以去除任何結垢。

- Reinstall level probe.
重新安裝水位探針。
- Once the level probe has been reinstalled, alert the trainer to inspect your work, then:
當水位探針已被重新安裝，請通知講師來檢查你的作業，然後：
 - Replace all machine panels.
重新安裝所有機械面板。

Skill B - Replace the burrs in an espresso grinder

技能 B-更換義式磨豆機的刀盤

- The necessary tools and replacement parts are at this station.
所需的工具及更換零件備妥在工作站中。
- Use the provided tools to
使用提供的工具完成下列事項：
 - Disassemble the grinder as needed to access the grinding burrs.
根據需求拆開磨豆機以檢查刀盤。
 - Remove the current set of grinding burrs, noting condition.
拆除目前的刀盤組，注意拆除的狀況。
Clean the grinding chamber and burr carriers.
清潔研磨腔室及刀座。
- Once the grinding chamber and burr carriers are sufficiently clean, alert the trainer to inspect your work, then:
當研磨腔室及刀座清潔乾淨時，請通知講師檢查你的作業，然後：
 - Install replacement grinding burrs.
安裝替換的刀盤。
 - Reassemble the grinder.
重新組裝磨豆機。
 - Connect the grinder to electrical power and test for proper function.
將磨豆機接上電源並測試功能是否正常。
 - Adjust to the approximately-correct setting
調整至大略正確的設定。

Scoring 計分

Place the resulting sub-total score for each requirement in the corresponding table on the Practical Exam Learner Worksheet.

將得出的小計總分填入實務測驗學員工作表的對應表格

Skill 3A - Clean a coffee batch brewer level probe

技能 3A- 清潔滴濾式咖啡機的水位探針

Task 任務	Points Possible 可能得分
Learner confirms that coffee brewer is disconnected from electrical power	2

學員確認滴濾式咖啡機未接上電源。	
Learner removes required access panel and places screws in a secure location 學員拆除所需的檢修孔蓋並將螺絲放置在妥善的位置。	2
Learner removes level probe without damaging probe or mounting bushings 學員在不損害探針或安裝套管的情況下拆除水位探針。	2
Learner uses appropriate means to fully clean the level probe 學員使用適當的方法完整地清潔水位探針。	2
Learner reinstalls level probe and connects wires if required 學員重新安裝水位探針並根據需求接上電線。	2
Learner reassembles access panel and tightens screws appropriately 學員重新安裝檢修孔蓋並適當地栓緊螺絲。	2
Learner records activity completion on Learner Worksheet 學員將活動完成情形記錄在學員工作表上。	1
Total 總分	13

Skill 3B - Replace the burrs in an espresso grinder

技能 3B-更換義式磨豆機的刀盤

Task 任務	可能得分 Points Possible
Learner unplugs espresso grinder before beginning service 學員在開始維護前拔除義式磨豆機的插頭。	2
Learner opens grinding chamber and removes worn grinding burrs 學員拆開研磨腔室並移除磨損的刀盤。	3
Learner thoroughly cleans grinding chamber and burr mounting surfaces 學員徹底地清潔研磨腔室及刀盤的安裝面。	2
Learner installs new grinder burrs, using an appropriate screw tightening procedure 學員使用合適的螺絲栓緊方式安裝新的刀盤。	3
Learner reassembles grinder and connects to electrical outlet	3

學員重新安裝磨豆機並接上插座。	
Learner tests for proper function and adjusts to approximately correct setting 學員測試功能是否正常並調整至大略正確的設定。	2
Learner records activity completion on Learner Worksheet 學員將活動完成情形記錄在學員工作表上。	1
Total 總分	16

Section 4 - Espresso Machine Preventive Maintenance

第四章-義式咖啡機的預防性維護

- **Total Time Available: 20 minutes for all three (3) parts.**
- **Points Available: 43, 30 points required to pass**
- **Silent working is required.**
- **This section is designed to evaluate the Learner's ability to:**
 - replace an espresso machine group screen and gasket
 - rebuild an espresso machine steam wand joint
 - record espresso machine service activity and inspection observations on a preventive maintenance checklist

測驗時間:總共 20 分鐘，分三個部份

得分:43 分，通過門檻為 30 分

作業中禁止交談

本章節旨在評估學員下列能力:

更換義式咖啡機的沖煮頭分水網及墊圈。

重新組裝義式咖啡機的蒸氣棒接頭。

在預防性維護核對表上記錄義式咖啡機維護活動及檢查觀測結果

Station Preparation 工作站前置準備

- Prepare a fully-operational espresso machine which **is connected to service**, powered on, and adjusted to manufacturer specifications.
準備一台能完整運作的義式咖啡機，並**接上服務**、電源且調整至符合製造商規範的設定。
- The espresso machine must utilize group screens which are retained by a screw.
義式咖啡機必須使用由螺絲固定的沖煮頭分水網。
- The espresso machine may utilize a ball style of steam wand joint or an alternative configuration.
義式咖啡機可以使用球形蒸氣棒接頭或替代配置。
- Create a space for individual work in this exam, either by erecting visual dividers between learners or placing them in separate rooms. One learner should not be able to see the work of a fellow learner.
利用在學員間設置視線擋板或是分配至獨立工作間以隔離出本測驗的個人工作空間。學員不應能夠看見鄰近學員的作業情況。

For Skill A

在技能 A

- Provide all necessary tools and supplies, including:
提供所有需要的器材，包含:
 - Group gasket
 - Group screen
 - Group screen screw
 - Screwdriver
 - Pick awl
 - Stiff-bristled brush沖煮頭墊圈

沖煮頭分水網
沖煮頭分水網螺絲
螺絲起子
尖錐
硬毛刷

For Skill B

在技能 B

- Provide all necessary tools and supplies, including:
提供所有需要的器材，包含：
 - Steam wand joint wear parts as-recommended by the espresso machine manufacturer
該義式咖啡機製造商建議的蒸氣棒接頭耗材
 - Appropriate lubricant
 - Appropriately-sized wrench(es), if required
 - Snap ring pliers, if required
 - Pick awl
 - Stiff-bristled brush
 - Steel wool or scrubbing pad
合適的潤滑劑
合適尺寸的扳手(如果有需求)
卡環鉗(如果有需求)
尖錐
硬毛刷
鋼絲絨或菜瓜布

For Skill C

在技能 C

- Provide a blank portafilter basket or thermofilter device for pump pressure observation
提供一個無孔濾杯或是測溫測壓把手以供觀察泵浦壓力。

Trainer Instructions 講師指示

- Review the Learner Instructions of the exercise with the Learner.
陪同學員詳讀學員測驗指示。
- Designate one brewing group and one steam wand joint for each Learner to service.
為各個學員指定一個沖煮頭及一只蒸氣棒接頭進行維護。
- Alert the Learner to begin and start the timer.
通知學員測驗開始並開始計時。
- If time expires prior to the completion of the task, the learner may receive partial credit for the work that has been completed.
假如時間在完成任務前結束，學員可以從已經完成的作業中得到部份分數。
- To receive full credit, the learner must perform the tasks correctly and mark all tasks correctly in the Learner Practical Exam Worksheet

為了得到滿分，學員必須正確執行任務並在學員實務測驗工作表上正確地標記所有任務

- This section includes working with tools. The learner must demonstrate safe working practices at all times during this section. Any unsafe actions will result in a score of 0 points for this section.

本章節包含工具操作。在測驗中學員必須隨時展現出安全的工作實踐。出現任何不安全的行為，本章節將會被評為 0 分。

Learner Instructions 學員指示

- There are three (3) parts to this section of the exam.
本章節測驗包含(3)個部分。
- Read the instructions for each part thoroughly.
請詳閱每個部份的說明。
- The trainer has provided a fully-operational espresso machine.
講師提供一台能完整運作的義式咖啡機。
- Service records for this machine indicate that it was last serviced one year prior to today's date
該設備的維護紀錄顯示最後維護日期為今日起算一年以前。
- In this section, you are to demonstrate a series of skills, detailed below, to complete preventive maintenance service on the provided espresso machine, including: a thorough inspection, servicing one brewing group, and servicing one steam wand joint.
在本章節中，你需要展示下文所述的一系列技能，來完成對提供的義式咖啡機的預防性維護服務，包含：一次徹底的檢查、維護一個沖煮頭以及維護一只蒸氣棒接頭。
- Record each task as complete on the Learner Practical Exam Worksheet, including observed values where required.
在學員實務測驗工作表上將各個任務記錄為完成，並記錄下要求的各項觀測到的數值。
- The trainer will tell you when your time begins.
當你的測驗時間開始時，講師將會告知你。
- Alert your trainer when you have completed this section.
當你完成這個本章節時請通知講師。
- **This section includes working with tools. Safe working practices must be demonstrated at all times during this section. Any unsafe actions will result in a score of 0 points for this section.**
本章節包含工具操作。在測驗中必須隨時展現出安全的工作實踐。任何不安全的行為將造成本章節被評為 0 分。

Skill A - Replace an espresso machine group screen and gasket

技能 A-更換一組義式咖啡機的沖煮頭分水網及墊圈

- The trainer has indicated which group is to be serviced
講師已指定需要維護的沖煮頭。
- The necessary tools and replacement parts are at this station.
需要的工具及更換零件備妥在本工作站。
- Use the provided tools to:
使用提供的工具進行下列事項:
 - Remove the group screen and gasket.
 - Clean the group.
 - Install a new group screen and gasket.
 - Test the brew group to check for leaks.

拆除沖煮頭分水網及墊圈。

清潔沖煮頭。

安裝新的沖煮頭分水網及墊圈。

測試沖煮頭是否漏水。

Skill B - Rebuild an espresso machine steam wand joint

技能 B-重新組裝一只義式咖啡機的蒸氣棒接頭

- The trainer has indicated which steam wand is to be serviced and which wear parts are to be replaced.
講師已經指定需要維護的蒸氣棒以及需要更換的耗材。
- The necessary tools and replacement parts are at this station.
需要的工具及更換零件備妥在本工作站。
- Use the provided tools to
使用提供的工具進行下列事項:
 - Remove the steam wand.
 - Remove the wear parts which require replacement.
 - Clean the steam wand.
 - Install new wear parts, as indicated, and reassemble the steam wand.
 - Test the steam wand to check for leaks.

拆除蒸氣棒。

拆除需要更換的耗材。

清潔蒸氣棒。

根據指示安裝新的耗材並重新組裝蒸氣棒。

測試蒸氣棒是否洩漏。

Skill C - Complete an espresso machine inspection

技能 C-完成義式咖啡機的檢查

- This section of the Practical Exam Learner Worksheet includes a Preventive Maintenance Inspection Checklist for this espresso machine
本章節的實務測驗學員工作表包含一份這台義式咖啡機的預防性維護檢查核對表。
- Inspect all of the items indicated and record your observations on your Learner Worksheet
檢查所有指定的物件並將觀察結果記錄在你的學員工作表上。

Scoring 計分

Place the resulting sub-total score for each requirement in the corresponding table on the Practical Exam Learner Worksheet.

將得出的小計總分填入實務測驗學員工作表的對應表格

Skill 4A - Replace an espresso machine group screen and gasket

技能 4A-更換一組義式咖啡機的沖煮頭分水網及墊圈

Task 任務	Points Possible 可能得分
Learner properly removes group screen and screw	2

學員正確地拆除沖煮頭分水網及螺絲。	
Learner properly removes group gasket without damaging brewing group 學員正確地拆除墊圈且未損壞沖煮頭。	3
Learner cleans group gasket channel and group 學員將墊圈溝槽及沖煮頭清潔乾淨。	2
Learner orients group gasket correctly when installing 學員以正確的一面定位墊圈。	2
Learner properly installs new group gasket 學員正確地安裝新墊圈。	2
Learner installs group screen and screw correctly, using appropriate screw torque 學員正確地安裝沖煮頭分水網及螺絲，並使用適當的力道拴緊螺絲。	2
Learner records activity completion on Learner Worksheet 學員將活動完成情形記錄在學員工作表上。	1
Total 總分	14

Skill 4B - Rebuild an espresso machine steam wand joint

技能 4B-重新組裝一只義式咖啡機的蒸氣棒接頭

Task 任務	Points Possible 可能得分
Learner properly removes steam wand from steam valve 學員正確地從蒸氣閥上拆下蒸氣棒。	2
Learner properly removes wear parts as directed without damage to equipment 學員根據引導正確地拆除磨損零件且未對設備造成損壞。	2
Learner cleans steam wand joint area and valve as directed 學員根據引導將蒸氣棒接頭區塊及蒸汽閥清潔乾淨。	2
Learner properly installs replacement parts, applying lubricant where applicable 學員正確地安裝更換零件，並在適當的地方添加潤滑劑。	3
Learner properly reassembles steam wand to steam valve 學員正確地將蒸氣棒重新安裝回蒸氣閥。	2
Steam wand joint does not leak when tested 確認測試時蒸氣棒接頭不會洩漏。	2

Learner records activity completion on Learner Worksheet 學員將活動完成情形記錄在學員工作表上。	1
Total 總分	14

Skill 4C - Complete an espresso machine inspection**技能 C-完成義式咖啡機的檢查**

Task 任務	Points Possible 可能得分
Learner correctly records steam boiler pressure on Learner Worksheet 學員正確地在學員工作表上記錄蒸氣鍋爐壓力。	1
Learner observes pump pressure and records on Learner Worksheet 學員觀測泵浦壓力並記錄在學員工作表上。	2
Learner inspects safety valve and records operating state on Learner Worksheet 學員檢查安全閥並將運作狀況記錄在學員工作表上。	1
Learner tests anti-vacuum valve and records operating state on Learner Worksheet 學員測試真空排除閥並將運作狀況記錄在學員工作表上。	2
Learner inspects expansion valve and records operating state on Learner Worksheet 學員檢查膨脹閥並將運作狀況記錄在學員工作表上。	1
Learner tests autofill system and steam boiler water level and records observations on Learner Worksheet 學員測試自動補水系統及蒸氣鍋爐水位並將觀測結果記錄在學員工作表上。	2
Learner tests drain function and records operating state on Learner Worksheet 學員測試排水功能並將運作狀況記錄在學員工作表上。	2
Learner inspects machine for water leaks and records status on Learner Worksheet 學員檢查機器是否漏水並將狀況記錄在學員工作表上。	2
Learner inspects machine for steam leaks and records status on Learner Worksheet 學員檢查機器的蒸氣是否會洩漏並將狀況記錄在學員工作表上。	2
Total 總分	15